9.3.4 Measurement of electrode potential

The overall redox reaction can be considered as the sum of two half reactions i.e., oxidation and reduction. Similarly, the emf of a cell can be considered as the sum of the electrode potentials at the cathode and anode,

$$E_{cell} = \left(E_{ox}\right)_{anode} + \left(E_{red}\right)_{cathode} \quad .....(9.19)$$

Here, $\left(E_{ox}\right)_{anode}$ represents the oxidation potential at anode and $\left(E_{red}\right)_{cathode}$ represents the reduction potential at cathode. It is impossible to measure the emf of a single electrode, but we can measure the potential difference between the two electrodes $\left(E_{cell}\right)$ using a voltmeter. If we know the emf of any one of the electrodes which constitute the cell, we can calculate the emf of the other electrode from the measured emf of the cell using the expression (9.19). Hence, we need a reference electrode whose emf is known

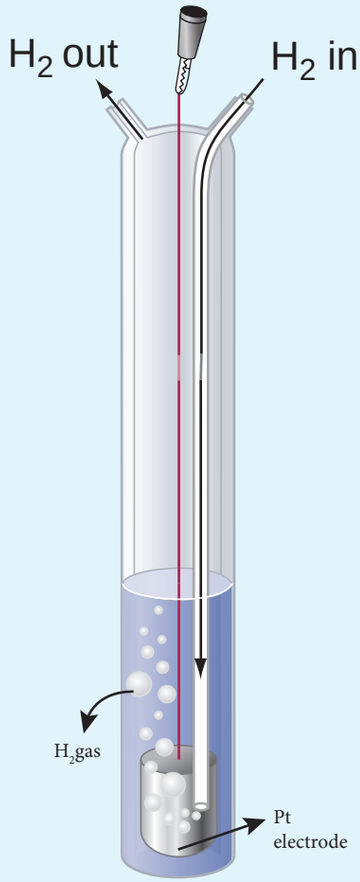
For that purpose, Standard Hydrogen Electrode (**SHE**) is used as the reference electrode. It has been assigned an arbitrary emf of exactly zero volt. It consists of a platinum electrode in contact with 1M HCl solution and 1 atm hydrogen gas. The hydrogen gas is bubbled through the solution at 25°C as shown in the figure 9.6. SHE can act as a cathode as well as an anode. The Half cell reactions are given below.

If SHE is used as a cathode, the reduction reaction is

$$2H^+(aq,1M)+2e^- \rightarrow H_2(g, 1\ atm) \qquad E^o = 0\ volt$$

If SHE is used as an anode, the oxidation reaction is

$$H_2(g,1\ atm) \rightarrow 2H^+(aq, 1M) + 2e^- \qquad E^o = 0\ volt$$



(SHE)

Figure 9.6 Standard Hydrogen electrode

Illustration

Let us calculate the reduction potential of zinc electrode dipped in zinc sulphate solution using SHE.

Step : 1 The following galvanic cell is constructed using SHE

$$Zn\ (s)\ |\ Zn^{2+}\ (aq, 1M)\ ||\ H^+\ (aq, 1M)\ |\ H_2\ (g, 1atm)\ |\ Pt\ (s)$$

Step : 2 The emf of the above galvanic cell is measured using a volt meter. In this case, the measured emf of the above galvanic cell is 0.76V.

Calculation

We know that,

$$E^\circ_{cell} = \left(E^\circ_{ox}\right)_{Zn|\,Zn^{2+}} + \left(E^\circ_{red}\right)_{SHE} \qquad [\text{From equation (9.19)}]$$

$E^\circ_{cell} = 0.76$ and $(E^\circ_{red})_{SHE} = 0V$. Substitute these values in the above equation

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